Report

By Rubab Fawad

In Silico Evaluation of Ligand Binding Affinity and Pharmacokinetic Properties Against 7SFB

Abstract

The integration of computational techniques in early-stage drug discovery enables rapid screening of potential therapeutic candidates. In this study, an in-silico workflow was employed to evaluate the binding affinity and pharmacokinetic properties of ten selected ligands against the protein 7SFB. Drug-likeness screening was conducted using Lipinski's Rule of Five, followed by ADMET profiling, binding site prediction, and molecular docking analysis. The results revealed that nine out of ten compounds satisfied drug-likeness criteria, indicating favorable oral bioavailability. Docking studies demonstrated binding energies ranging from −4.7 to −6.7 kcal/mol, with Metformin exhibiting the highest binding affinity. These findings highlight Metformin as a promising candidate for further investigation. Overall, the study demonstrates the effectiveness of integrated computational approaches in identifying potential drug candidates.

1. Introduction

The drug discovery process is traditionally time-consuming and resource-intensive. Computational approaches, including molecular docking and pharmacokinetic prediction, have significantly enhanced the efficiency of identifying promising drug candidates. These methods allow rapid screening of compounds and provide insights into ligand–protein interactions prior to experimental validation.

The protein 7SFB was selected as the target due to its well-resolved three-dimensional structure and suitability for computational analysis. The ability to evaluate ligand binding within a defined active site makes it an appropriate model for studying molecular interactions.

Materials and Methods

2. Ligand Selection and Preparation

Ten ligands Aspirin, Ibuprofen, Acetaminophen, Naproxen, Diclofenac, Caffeine, Theophylline, Metformin, Artemether, and Lumefantrine were retrieved from PubChem. The selected compounds represent structurally diverse molecules with known pharmacological relevance. Ligands were prepared and converted into appropriate formats using Open Babel for compatibility with docking software.

3. Drug-likeness Screening

The smiles of the given ligand: Aspirin, Ibuprofen , Acetaminophen, Naproxen, Diclofenac, Caffeine, Theophylline, Metformin, Lopinavir, Paclitaxel were copied from PubChem and pasted in a table.

Table1: shows the compounds given, their PubChem CID numbers and smiles from PubChem.

No#	Compound Name	PubChem CID	Smiles
01.	Aspirin	2244	CC(=O)OC1=CC=CC=C1C(=O)O
02.	Ibuprofen	3672	CC(C)CC1=CC=C(C=C1)C(C)C(=O)O
03.	Acetaminophen	1983	CC(=O)NC1=CC=C(C=C1)O
04.	Naproxen	156391	C[C@@H](C1=CC2=C(C=C1)C=C(C=C OC)C(=O)O
05.	Diclofenac	3033	C1=CC=C(C(=C1)CC(=O)O)NC2=C(C= C=C2Cl)Cl
06.	Caffeine	2519	CN1C=NC2=C1C(=O)N(C(=O)N2C)C
07.	Theophylline	2153	CN1C2=C(C(=O)N(C1=O)C)NC=N2
08.	Metformin	4091	CN(C)C(=N)N=C(N)N
09.	Artemether	68911	C[C@@H]1CC[C@H]2[C@H]([C@H](C @H]3[C@@]24[C@H]1CC[C@](O3)(O)C)OC)C
10.	Lumefantrine	6437380	CCCCN(CCCC)CC(C1=CC(=CC\\2=C1 =C(/C2=C/C4=CC=C(C=C4)Cl)C=C(C=)Cl)Cl)O

Drug Likeliness screening of the given ligand was done using SwissADME website and the Lipinski was checked for each compound. Lipinski rule of (Ro5) is widely used rule in medicinal chemistry that governs whether it is orally active drug in humans. Lipinski's Rule checks:

- Molecular weight ≤ 500
- LogP ≤ 5
- Hydrogen bond donors ≤ 5
- Hydrogen bond acceptors ≤ 10

If a compound follows these better absorption and better oral bioavailability.

Table2: shows compounds given and their drug-likeliness (Lipinski) score with SwissADME.

No#	Compound name	PubChem CID	Drug-likeliness (Lipinski rule of 5)
01.	Aspirin	2244	Yes; 0 violation
02.	Ibuprofen	3672	Yes; 0 violation
03.	Acetaminophen	1983	Yes; 0 violation
04.	Naproxen	156391	Yes; 0 violation
05.	Diclofenac	3033	Yes; 0 violation
06.	Caffeine	2519	Yes; 0 violation
07.	Theophylline	2153	Yes; 0 violation
08.	Metformin	4091	Yes; 0 violation
09.	Artemether	68911	Yes; 0 violation
10.	Lumefantrine	6437380	No; 2 violations

Interpretation: Out of 10 compounds, 9 compounds obey Lipinski's Rule of Five (0 violations) and have good oral drug-like properties, making them suitable candidates for further development. However, one compound (Lumefantrine) shows 2 violations.

3.1 Strong Drug-Like Compounds (0 violations)

Aspirin, Ibuprofen, Acetaminophen, Naproxen, Diclofenac, Caffeine, Theophylline, Metformin, Artemether show zero violation, making them pharmacokinetically favorable and likely to be well absorbed orally.

Exception: Lumefantrine (2 violations)

Although Lumefantrine shows good binding affinity (−6.1 kcal/mol), its Lipinski violations suggest potential limitations in oral bioavailability and absorption due to its High molecular weight and High lipophilicity (LogP).

Key Scientific Insight

Even through drug-likeness alone does not guarantee strong binding, and strong binding does not ensure good pharmacokinetics. Combining Lipinski screening with docking is essential for identifying viable drug candidates.

3.2 Discussion

All selected ligands were evaluated for drug-likeness using Lipinski's Rule of Five via SwissADME. The results indicated that nine out of ten compounds exhibited zero violations, suggesting favorable oral bioavailability and pharmacokinetic properties.

Lumefantrine was the only compound that showed two violations, due to its higher molecular weight and lipophilicity, which may limit its absorption.

Notably, Metformin, which demonstrated the highest binding affinity in docking studies, also satisfied all Lipinski criteria, reinforcing its potential as a promising candidate. These findings highlight the importance of integrating drug-likeness screening with molecular docking to identify compounds with both strong binding potential and suitable pharmacokinetic profiles.

4.Protein Preparation and cleaning

Uncleaned protein ***7SFB,*** also known as *Main Protease,* is an essential constructive part of the COVID-19 virus without which the virus is rendered from replication. This protein is the master key to the virus survival. This protein 7SFB 's PDB structure was taken from the RCSB PDB website and was then visualized and cleaned via PYMOL.

- **Method:** X-RAY DIFFRACTION
- **Resolution:** 1.90 Å

7SFB is a viral protein the consists of sequence of 306 amino acids with unique protein chain A .

4.1 Visualized in PYMOL:

This structure was then visualized in PYMOL .

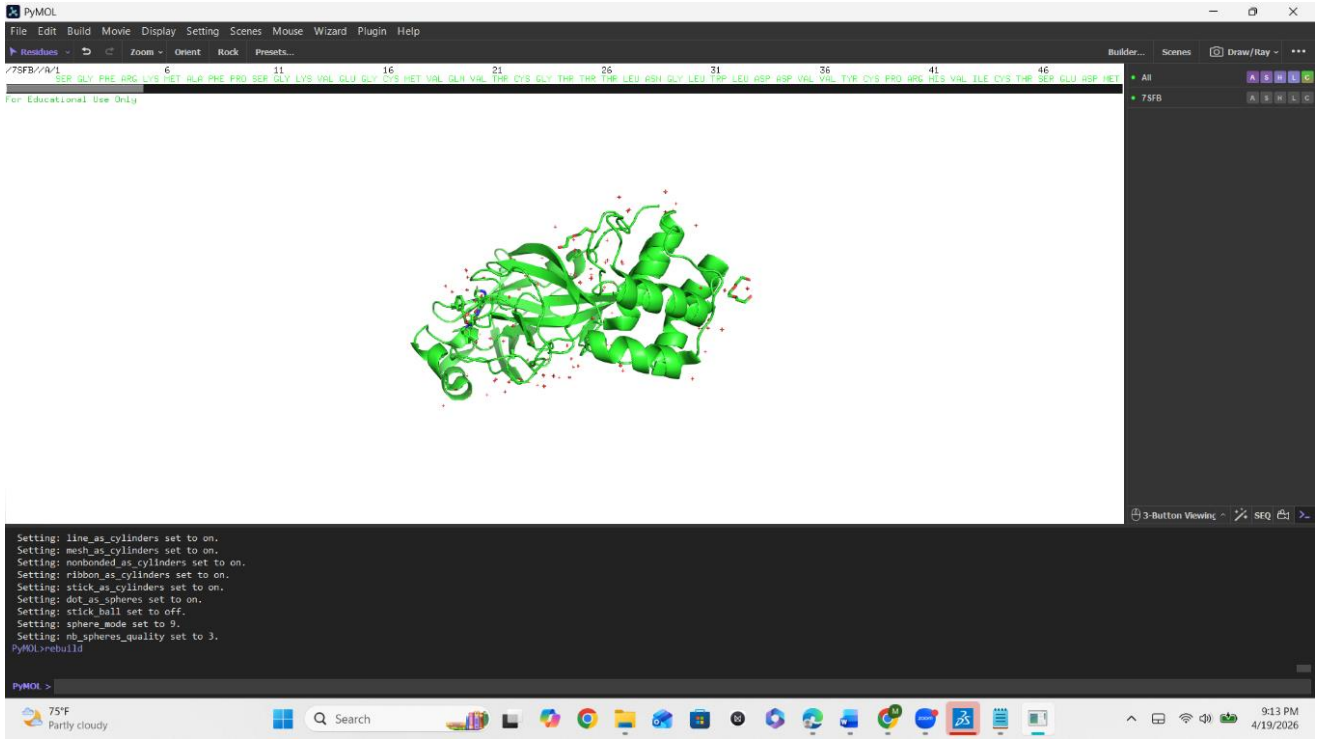


Figure 1 shows the uncleaned protein taken from the RCSB PDB website visualized in PYMOL

4.2 Protein Cleaning in Chimera:

7SFB was opened in Chimera and then.

1)select → residue → non-essential amino acids/HOH

2) actions → atoms/bonds → delete

3) Tool → structure Editing → Dock prep

Save this file as 7SFB_cleaned.pdb.

Visualization in Chimera:

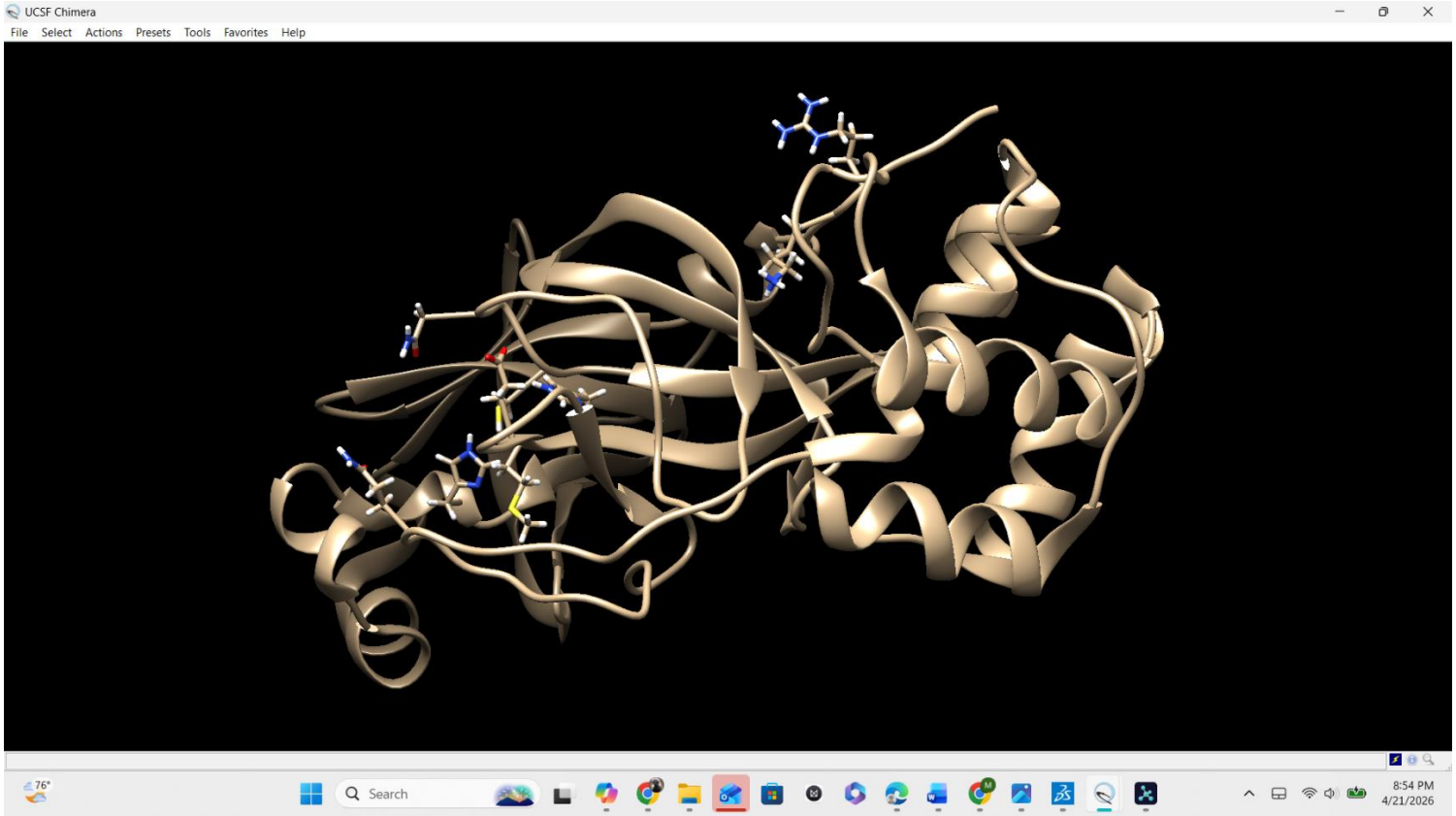


Figure 2 shows the cleaned protein in chimera

4.3 Visualization in PYMOL:

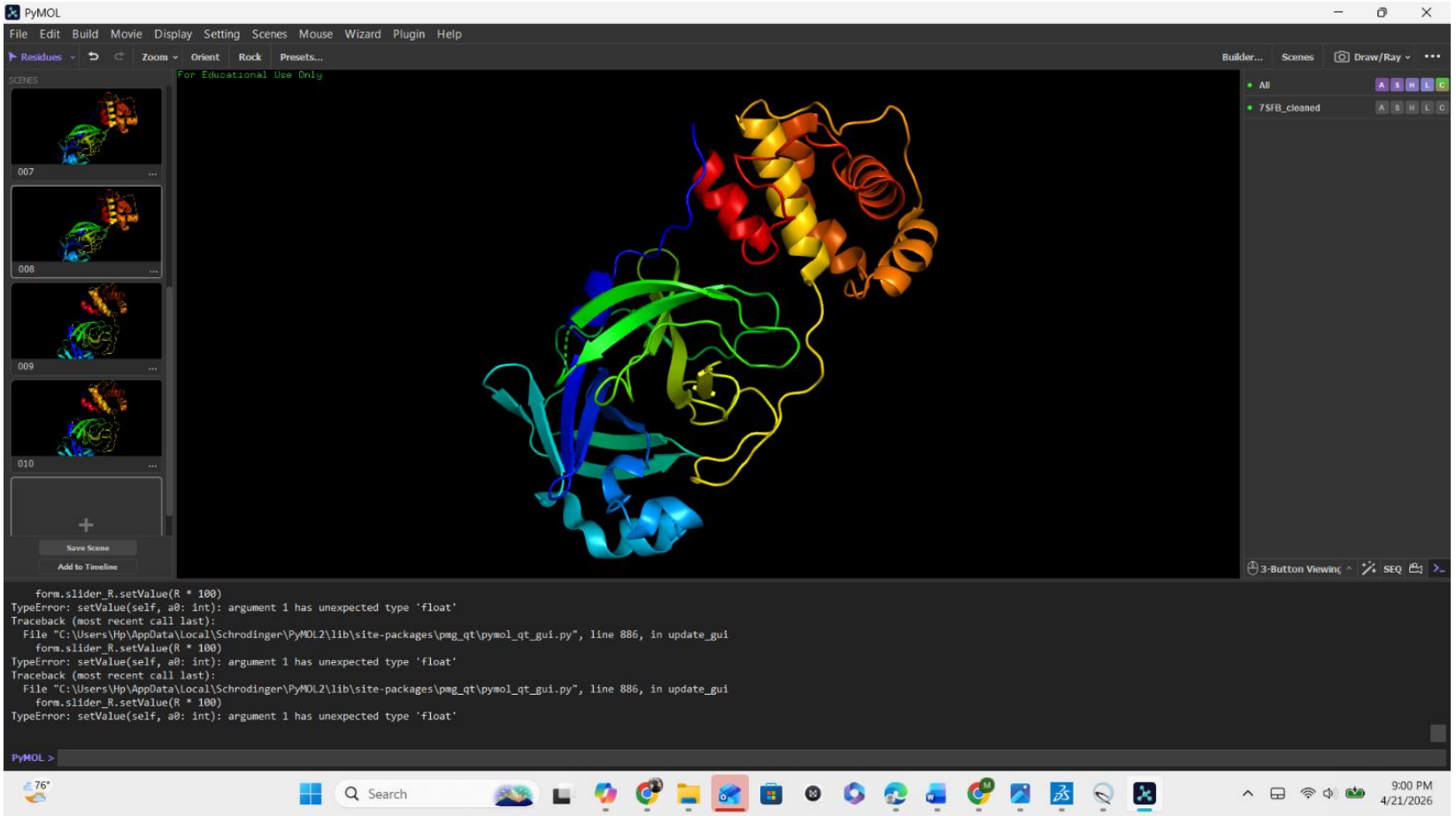


Figure 3 the protein cleaned was then visualized in PYMOL

5.DoGSiteScorer : Binding Site Prediction

The binding site of protein 7SFB was predicted by DoGSiteScorer where uncleaned/raw protein was uploaded and binding site was analyzed. The visuals are shown below:

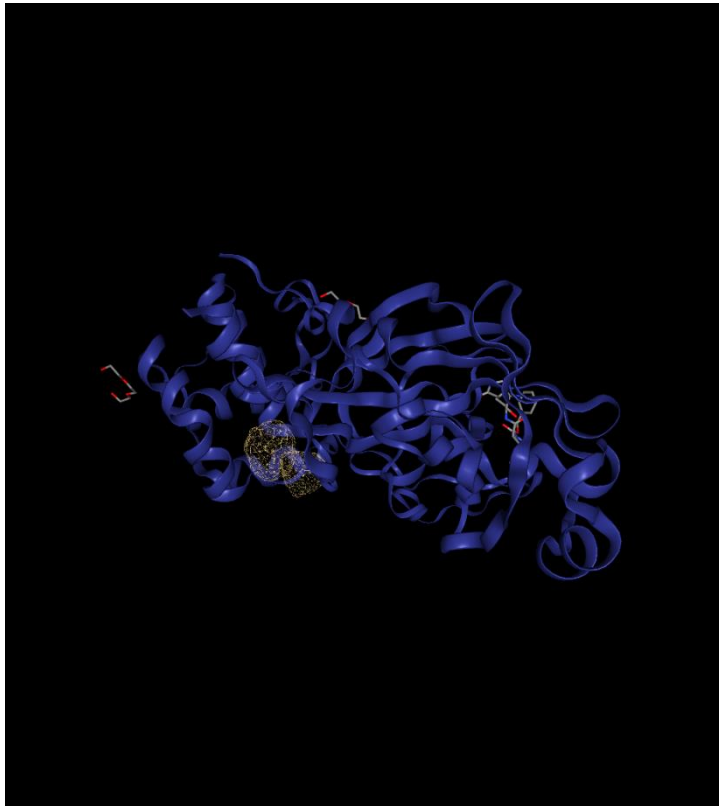


Figure 4 shows closer look to the binding site P_0

Figure 5 shows the binding site P_0

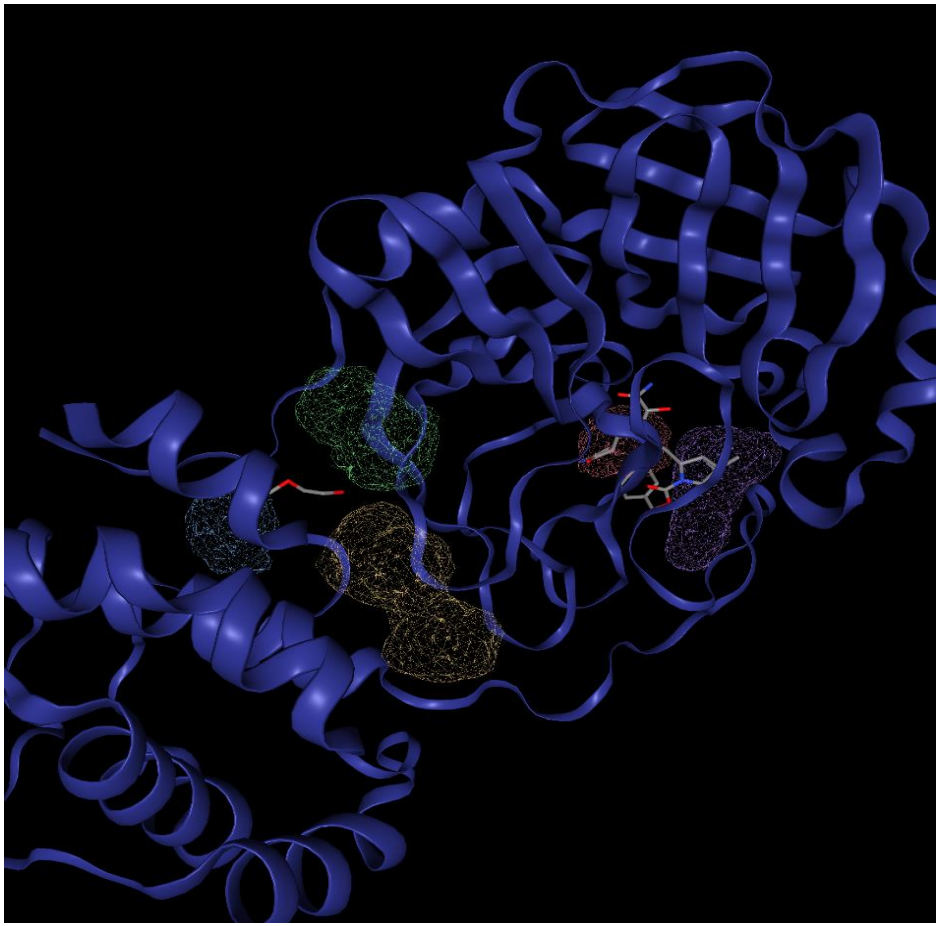


Figure 6 shows five best binding sites P_0 to P_4

5.1 Visualizing Amino acid Residues in Binding Site in PYMOL

The binding site results obtained from DoGSiteScorer were extracted and P_1 residues sequence with highest volume [Å³]=156.672, surface [Å²]=398.806, depth [Å]=10.4 was selected and visualized in PYMOL. The sequence pf amino acid residues were noted for docking.

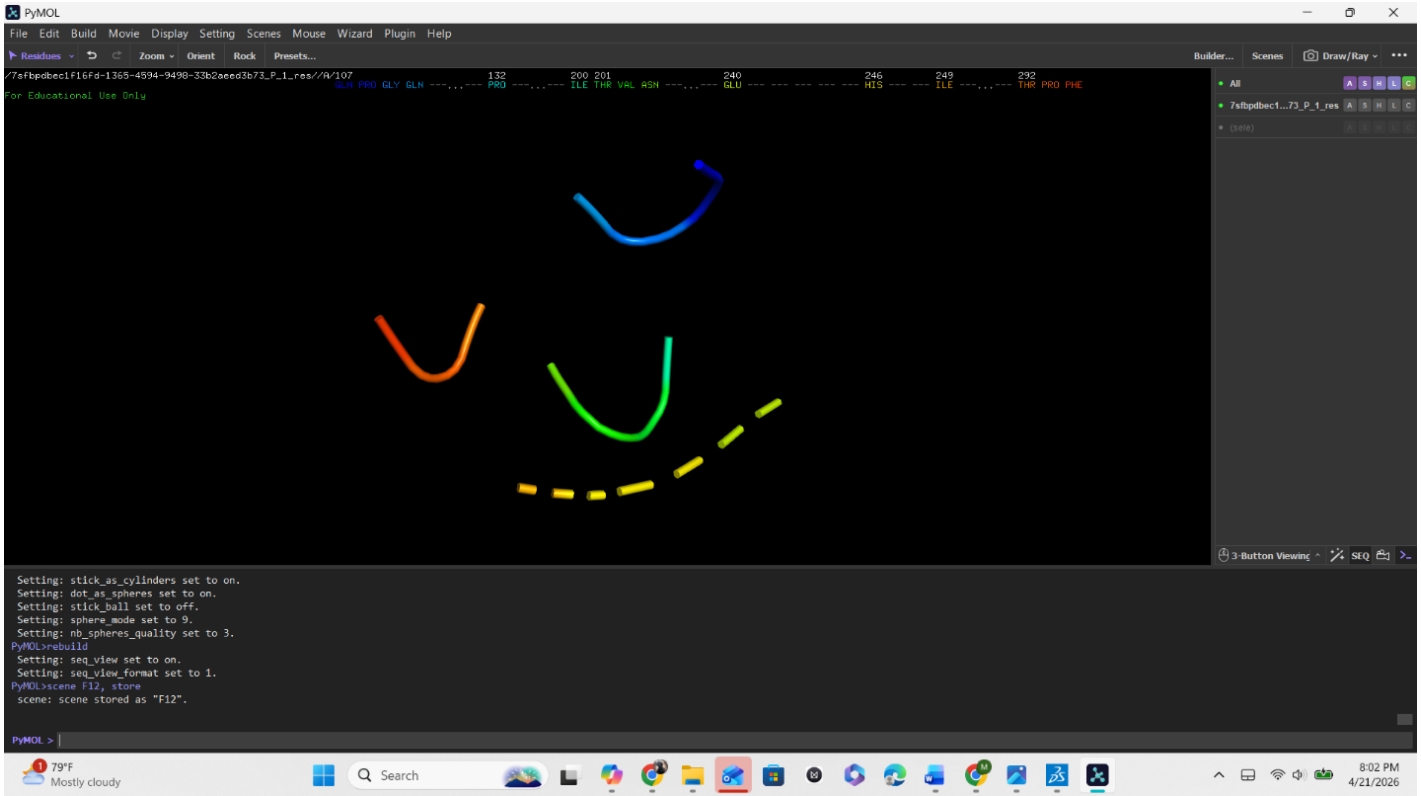


Figure 7 shows the amino acid residues sequence obtained from binding site prediction in DoGSiteSorer

The Amino acid Sequence noticed is:

107	108	109	110	132	200	201	202	203	240	246	249	292	293	294
GLN	PRO	GLY	GLN	PRO	ILE	THR	VAL	ASN	GLU	HIS	ILE	THR	PRO	PHE

6.Docking: Binding site prediction

The binding affinity of the target protein 7SFB was determined with the given ligands in PYRX. Cleaned protein structure was loaded and converted to the macromolecule (pdbqt). After that, all the ligand were loaded one by one through open babel and binding site residue sequence was selected which was previously obtained through the DoGSiteScorer website. The docking was performed via Vina, and the grid box was extended to cover all the residues. Docking was performed and the binding affinities of different ligand were observed for the target protein 7SFB. Shown in the table below:

Table3: shows the ligand, their PubChem CID, docking score and interpretation.

No#	Compound Name	PubChem CID	Binding Energy kcal/mol	Interpretation
01.	Aspirin	2244	-4.8	weakest binding interactions
02.	Ibuprofen	3672	-5.6	moderate binding affinity
03.	Acetaminophen	1983	-5.3	moderate binding affinity
04.	Naproxen	156391	-5.5	moderate binding affinity
05.	**Diclofenac**	**3033**	**-6.5**	strong binding affinity
06.	Caffeine	2519	-6	moderate–strong interactions
07.	Theophylline	2153	-4.7	weakest binding interactions
08.	**Metformin**	**4091**	**-6.7**	strongest binding affinity

09.	Artemether	68911	-6.3	strong binding affinity
10.	Lumefantrine	6437380	-6.1	moderate–strong interactions

6.1 Interpretation

The docking results show that binding energies range from −4.7 to −6.7 kcal/mol, indicating ligand–protein interactions. Among all the compounds, Metformin showed the strongest binding affinity (−6.7 kcal/mol), followed by Diclofenac (−6.5 kcal/mol) and Artemether (−6.3 kcal/mol), suggesting stable and favorable interactions within the active site.

Caffeine and Lumefantrine showed moderate binding potential, while Ibuprofen, Naproxen, and Acetaminophen demonstrated moderate affinity. Aspirin and Theophylline exhibited the weakest interactions.

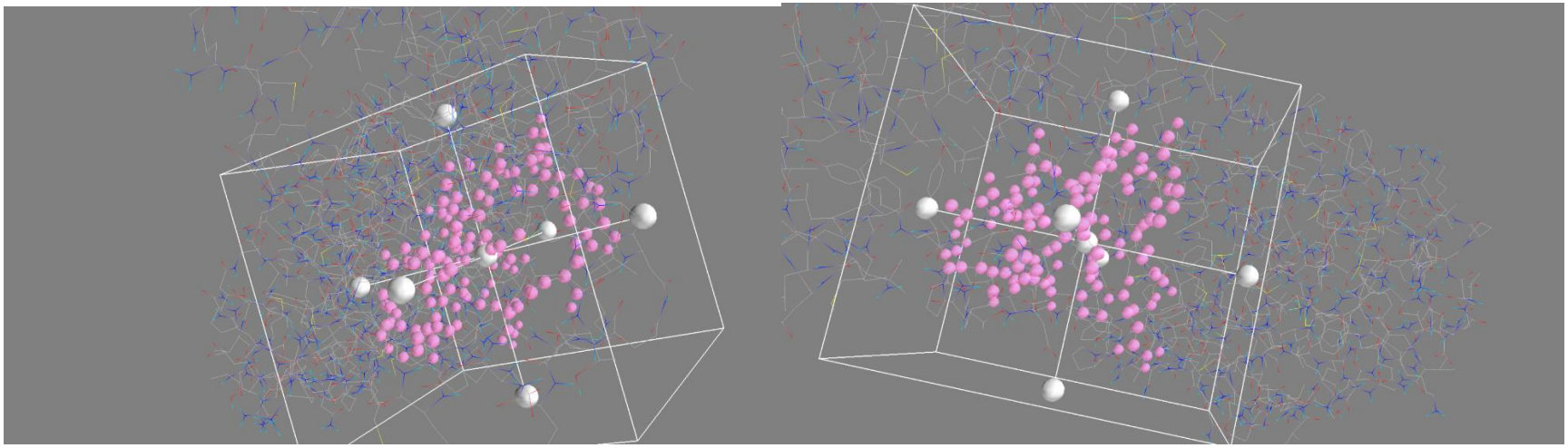


Figure 8 shows the grid box covering all the residue amino acids in binding site to predict docking score

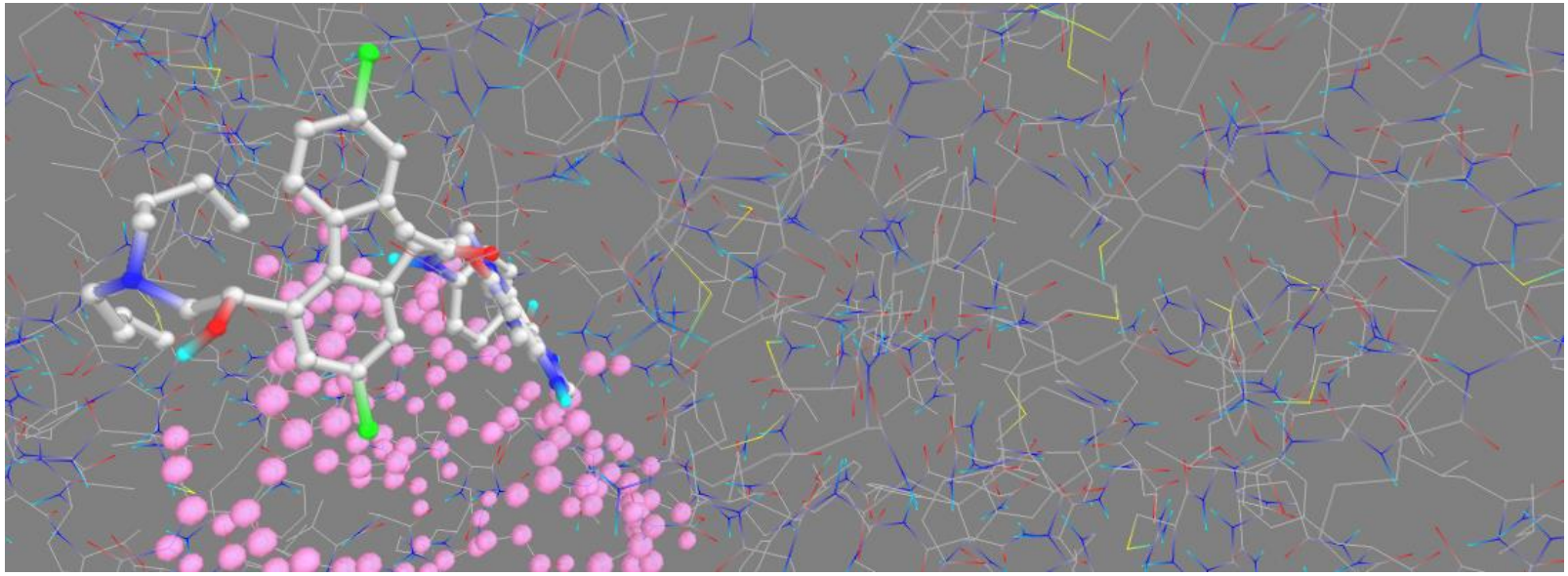
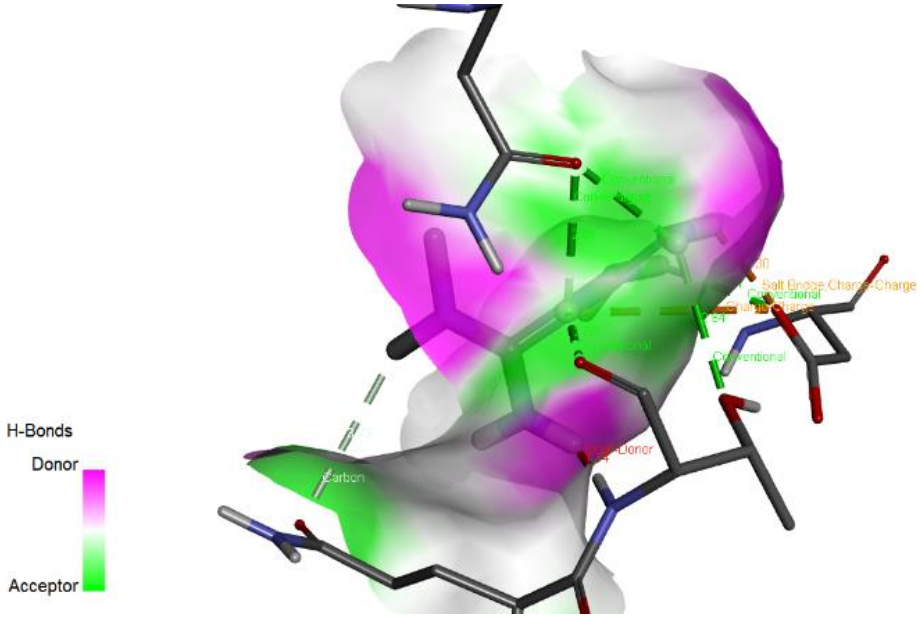


Figure 9 shows the binding of ligand to the residue amino acids of the protein

6.2 Visualization via Discovery Studio:

The protein 7SFB_cleaned.pdbqt and its interactions with the ligand with highest binding affinity Metformin, diclofenac, Artemether, Lumefantrine, and caffeine were observed.

6.2.1 Metformin



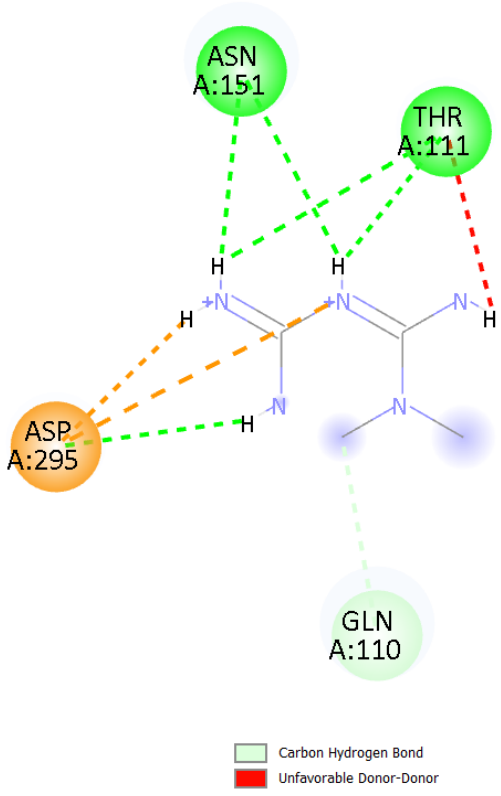


Figure 10 shows the 3D depiction of Hydrogen bonding of the ligand with the amino acids (right) , left shows the interactions of Metformin to the amino acids of protein

6.2.2 Diclofenac

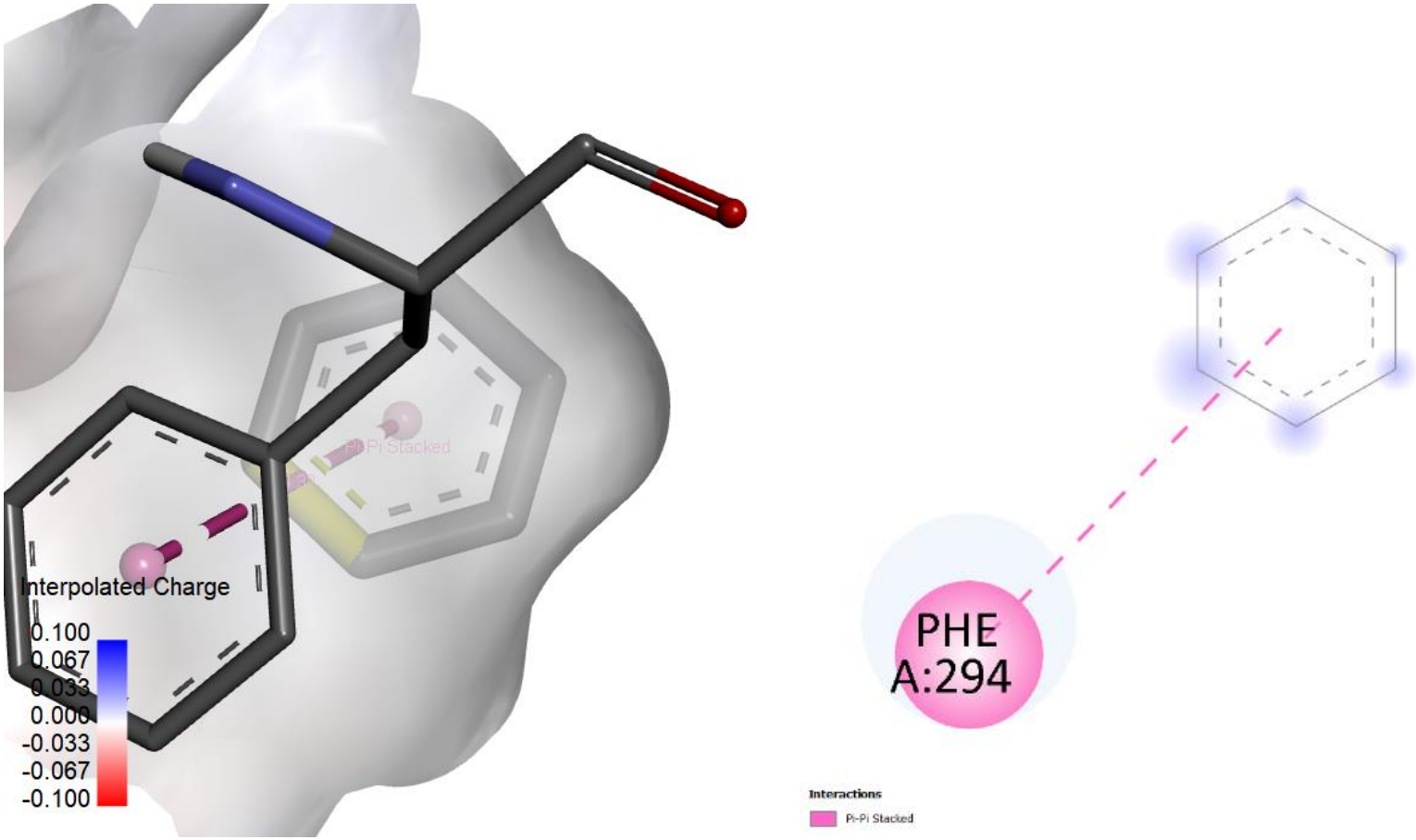


Figure 11 left side shows the interpolated charge while the right site shows 3D picture of the intermolecular interactions between Diclofenac aromatic ring and amino acid PHE294

6.2.3 Artemether

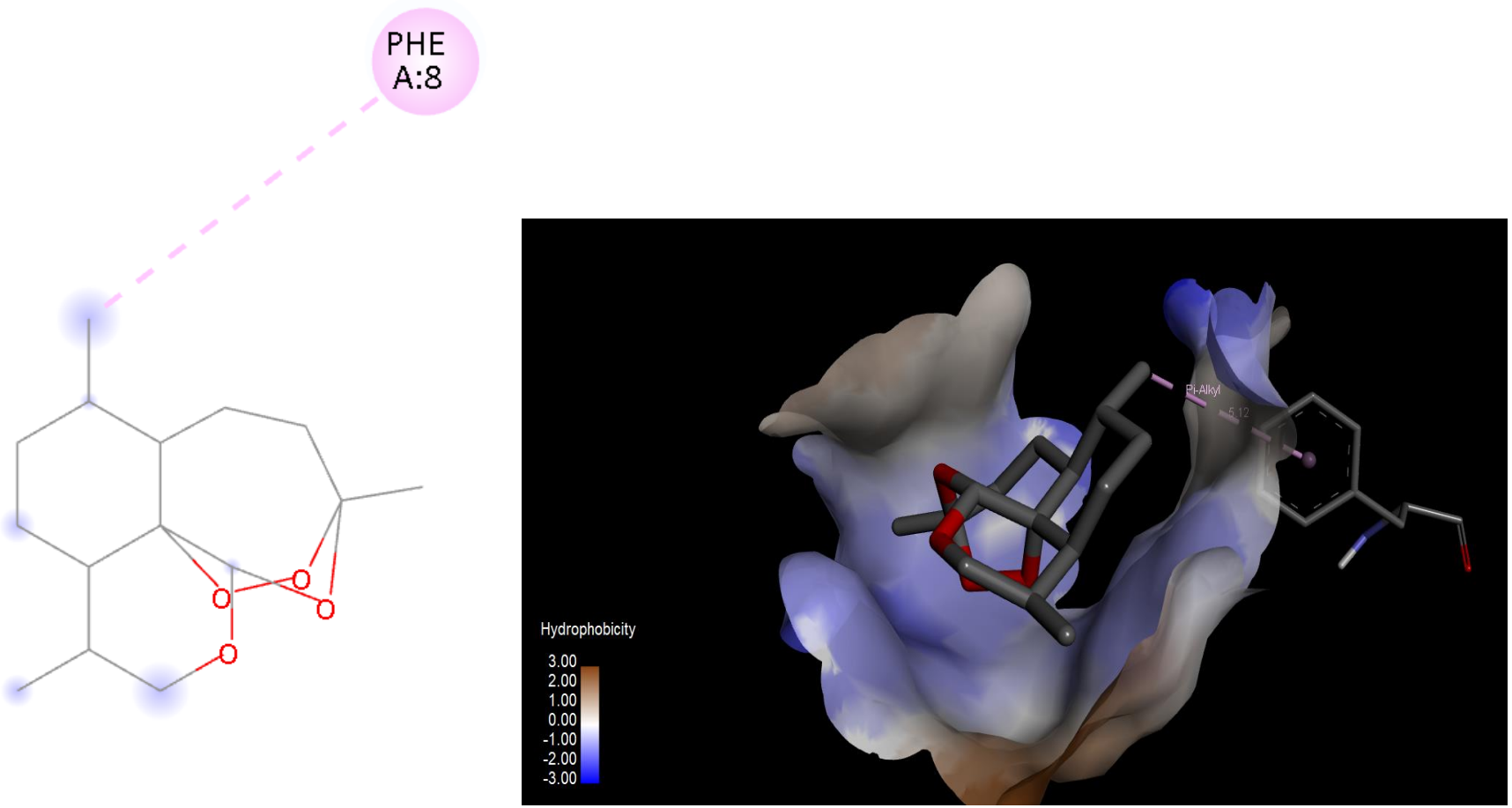


Figure 12 left side shows the the interactions of artemether with the PHEa:8 amino acid while the left side shows 3D interactions and hydrophobicity

6.2.4 Lumefantrine

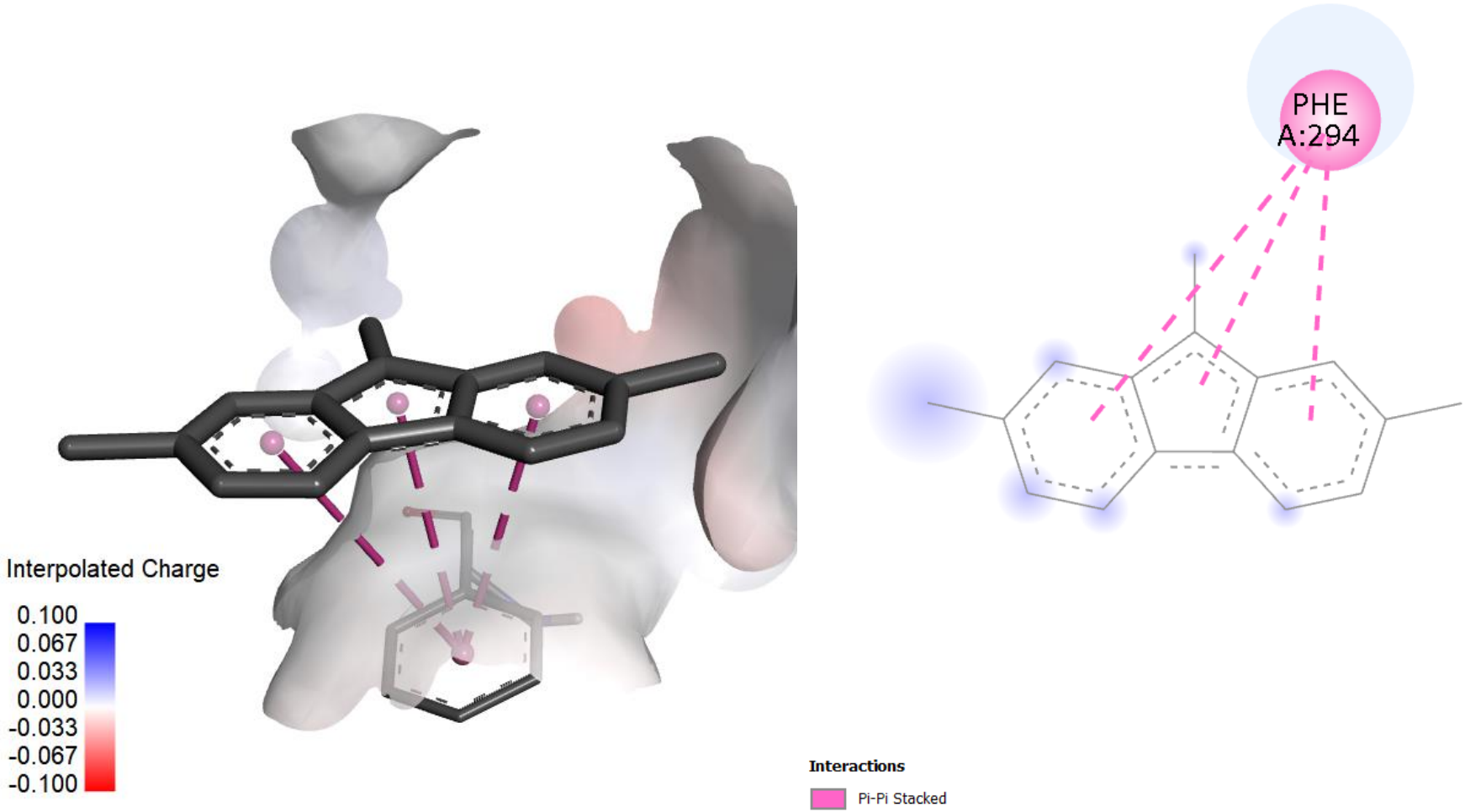


Figure 13 left side shows the 3d interactions of Lumefantrine with protein amino acid and interpolated charge, left side emphasizes the 2D picture of interactions.

6.2.5 Caffeine

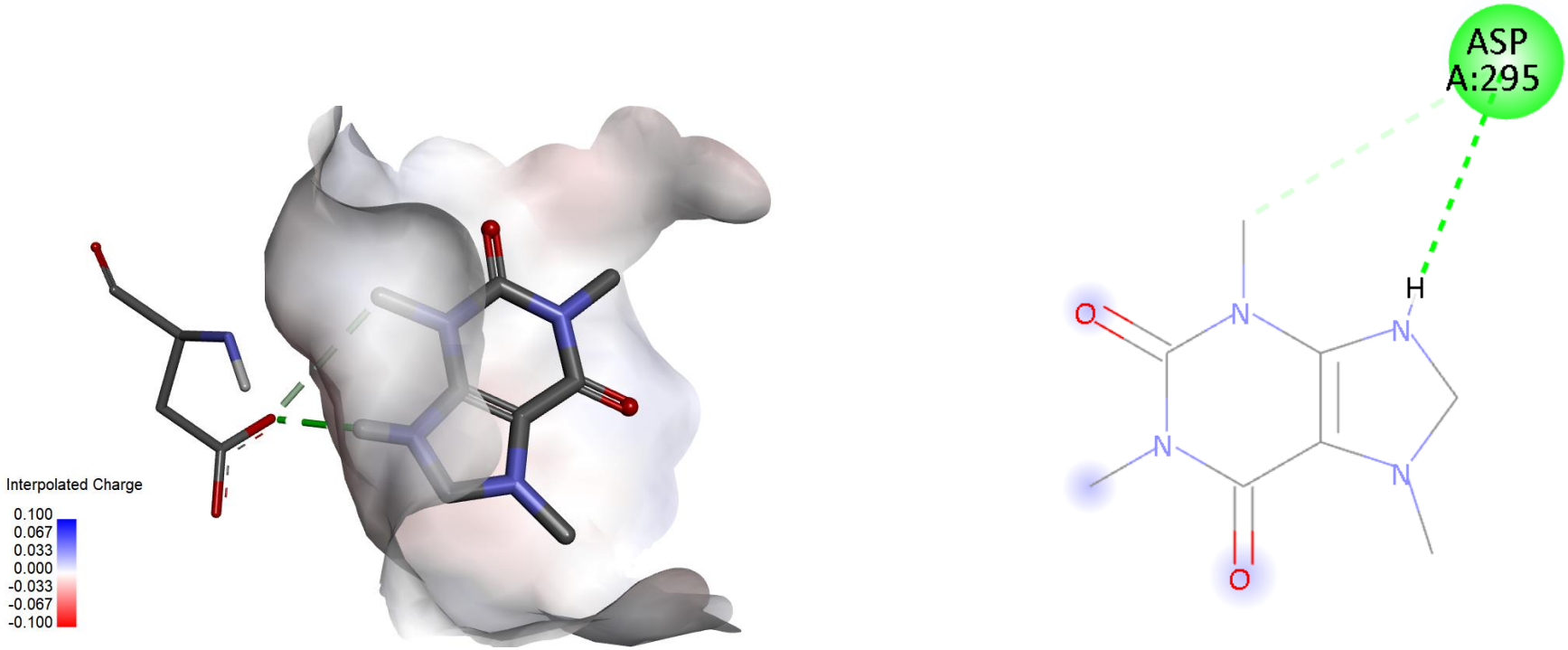


Figure 14 shows the 3D interactions between protein and the caffeine molecule with interpolated charge cloud , while the left side explores the 2D interactions of Caffeine with the protein amino acid ASP of chain A at position 295.

7. ADMET prediction

F30% / F50%Bioavailability:

The percentage of the drug that successfully reaches the bloodstream.

Caco-2Absorption:

How well the drug passes through the intestinal wall (more positive is better).

BBB Access:

Whether the drug can cross the Blood-Brain Barrier to affect the Central Nervous System.

Fu Free Fraction:

The % of the drug NOT stuck to blood proteins; only "free" drug is active.

HLM Stability Metabolic Speed:

How quickly does the liver break down the drug (High stability = stays in body longer).

CYP Substrate Interactions:

Which liver enzymes process the drug (determines risk of drug-drug interactions).

$T_{1/2}$ Half-Life:

The time it takes for the drug concentration to reduce by half.

Toxicity Scores Safety:

Likelihood of irritation or long-term harm (1.0 is high risk, 0.0 is low).

Table4: Shows ADMET properties of the Compounds.

No#	Compound Name	Absorption	Distribution	Metabolism	Excretion	Toxicity
01.	Aspirin	-F30% excellent -F50% excellent -Caco-2 Permeability 4.985	-BBB +++ excellent -Fu=40.1% -OATP1B1 inhib excellent	-HLM Stability ~poor - CYP3A4 substrate -- -CYP2B6 substrate --	-CL_{plasma} =2.766 -T1/2 =0.777	skin Sensitization=0.705! Carcinogenicity=0.2 ok Eye Corrosion=0.919 !
02.	Ibuprofen	-F30% excellent -F50% excellent -Caco-2 Permeabilit 4.301	-BBB + medium -Fu=2.6% ! -OATP1B1inhibitor medium	-HLM Stability ! - CYP3A4 substrate ++ -CYP2B6 substrate ---	-CL_{plasma} = 1.038 -T1/2 =1.4 excellent	skin sanitization =0.983 -Carcinogenicity=0.518 Eye Corrosion=0.952
03.	Acetaminoph	-F30% excellent -F50% excellent -Caco-2 Permeability 4.272	-BBB medium -Fu=65.6% -OATP1B1 inhibitor poor	-HLM Stability = poor - CYP3A4 substrate - -CYP2B6 substrate ---	-CL_{plasma} =6.178 -T1/2 =5.088	-skin sanitization=0.799 ! -Carcinogenicity=0.135 excellent -Eye Corrosion=0 medium
04.	Naproxen	-F30% excellent -F50% excellent -Caco-2 Permeabilit 4.11	-BBB excellent -Fu=30% -OATP1B1 inhibitor medium	-HLM Stability++ - CYP3A4 substrate ++ -CYP2B6 substrate ---	-CL_{plasma} =8.903 -T1/2 =1.115	-skin sanitization =0.893! -Carcinogenicity=0.558 Eye Corrosion=0.354
05.	Diclofenac	-F30% excellent -F50% excellent	-BBB excellent -Fu=1.9% -OATP1B1inhibitor medium	-HLM Stability ! - CYP3A4 substrate --- -CYP2B6 substrate ---	-CL_{plasma} = 0.375 -T1/2 =1.786	skin sanitization =0.295 -Carcinogenicity=0.188 Eye Corrosion=0.017

		-Caco-2 Permeabilit 4.657				
06.	Caffeine	-F30% excellent -F50% excellent -Caco-2 Permeabilit 4.303	-BBB +++ ! -Fu=62.8% -OATP1B1 inhibitor medium	-HLM Stability --- poor - CYP3A4 substrate ++ -CYP2B6 substrate ---	-CL_{plasma} =0.909 -T1/2 =1.976	-skin sanitization =0.393 -Carcinogenicity=0.498 -Eye Corrosion=0.072
07.	Theophylline	-F30% excellent -F50% excellent -Caco-2 Permeabilit 4.301	-BBB +++ ! -Fu=62.8% -OATP1B1 inhibitor medium	-HLM Stability ! - CYP3A4 substrate ++ -CYP2B6 substrate+++	-CL_{plasma} =0.909 -T1/2 =1.976	-skin sanitization =0.393 -Carcinogenicity=0.498 Eye Corrosion=0.117
08.	Metformin	-F30% excellent -F50% excellent -Caco-2 Permeabilit 5.33	-BBB + medium -Fu=105.7 % -OATP1B1 inhibitor excellent	-HLM Stability ! - CYP3A4 substrate ++ -CYP2B6 substrate ---	-CL_{plasma} = 6.3 -T1/2 = 1.897	skin sanitization =0.696 -Carcinogenicity=0.636 Eye Corrosion=0.022
09.	Artemether	-F30% excellent -F50% excellent -Caco-2 Permeabilit 4.847	-BBB --- excellent -Fu=22.5% -OATP1B1 inhibitor + poor	-HLM Stability + excellent - CYP3A4 substrate - -CYP2B6 substrate ++	-CL_{plasma}=13.876 -T1/2 =1.916	-skin sanitization =0.999! -Carcinogenicity=0.133 -Eye Corrosion=0.047
10.	Lumefantrine	-F30% excellent -F50% excellent	-BBB +++ poor -Fu=0.8% !	-HLM Stability+++ - CYP3A4 substrate ++ -CYP2B6 substrate ---	-CL_{plasma} = 5.888 -T1/2 =0.68	-skin sanitization =0.997! -Carcinogenicity=0.112 Eye Corrosion=0.001

		-Caco-2 Permeabilit 5.088	-OATP1B1 inhibitor + Poor			

7.1 Interpretation of the Data

7.1.1. Absorption (The "Entry")

- All compounds (Aspirin to Lumefantrine) show **Excellent F30/50%**, meaning they are highly effective when taken orally.
- **Metformin** has the lowest Caco-2 permeability (-5.33), meaning it struggles to pass through membranes.

7.12. Distribution (The "Travel")

- **Brain Reach:** Aspirin, Caffeine, and Theophylline show "excellent" BBB penetration.
- **Protein Binding: Lumefantrine** (0.8%) and **Diclofenac** (1.9%) have very low **Fu**. This means 98-99% of the drug is "trapped" on proteins in the blood and acts as a reservoir.

7.1.3. Metabolism (The "Processing")

- **Liver Load: Artemether** and **Lumefantrine** show "excellent" HLM stability, meaning the liver does not destroy them immediately.
- **Interaction Risk: Theophylline** is a substrate for both **CYP3A4** and **CYP2B6**. This makes it the most "sensitive" drug on the list for potential interactions with other medications.

7.1.4. Excretion (The "Exit")

- **Duration: Acetaminophen** has a high half-life (5.088), meaning it lingers much longer than **Lumefantrine** (0.68), which is cleared very rapidly.

- **Clearance: Artemether** has the highest CL_{plasma} (13.876), showing the body works hardest to pump this specific compound out.

7.1.5. Toxicity (The "Risk")

- **Irritants: Artemether** and **Lumefantrine** have dangerously high skin sensitization scores (0.99+), suggesting they are highly likely to cause allergic skin reactions.
- **Safety: Diclofenac** and **Lumefantrine** have zero Eye Corrosion risk (0.01 and 0.001), making them "cleaner" in that specific category than **Aspirin** (0.919).
- **Cancer Risk: Acetaminophen** and **Artemether** show the lowest carcinogenicity scores (0.13), indicating a very high safety profile for long-term use.

Best among all : Diclofenac and Acetaminophen

8. Alphafold3 prediction of the protein, and comparison with the cleaned protein.

The backbone and atom positions of AlphaFold 3 model and cleaned crystal structure (7SFB) are almost identical. The structure was obtained from AI AlphaFold3 website by providing FASTA sequence of amino acids and then it predicted the right structure with high confidence of RMSD of 0.616 angstrum.

0 - 2 Å → Excellent match → High confidence.

2 - 5 Å → Moderate match → Significant loop or side-chain differences.

> 10 Å → Structural Mismatch → A **domain swap**, **misalignment**, or **different conformational state**.

Result: RMSD = 0.616 Å shows and outstanding match.

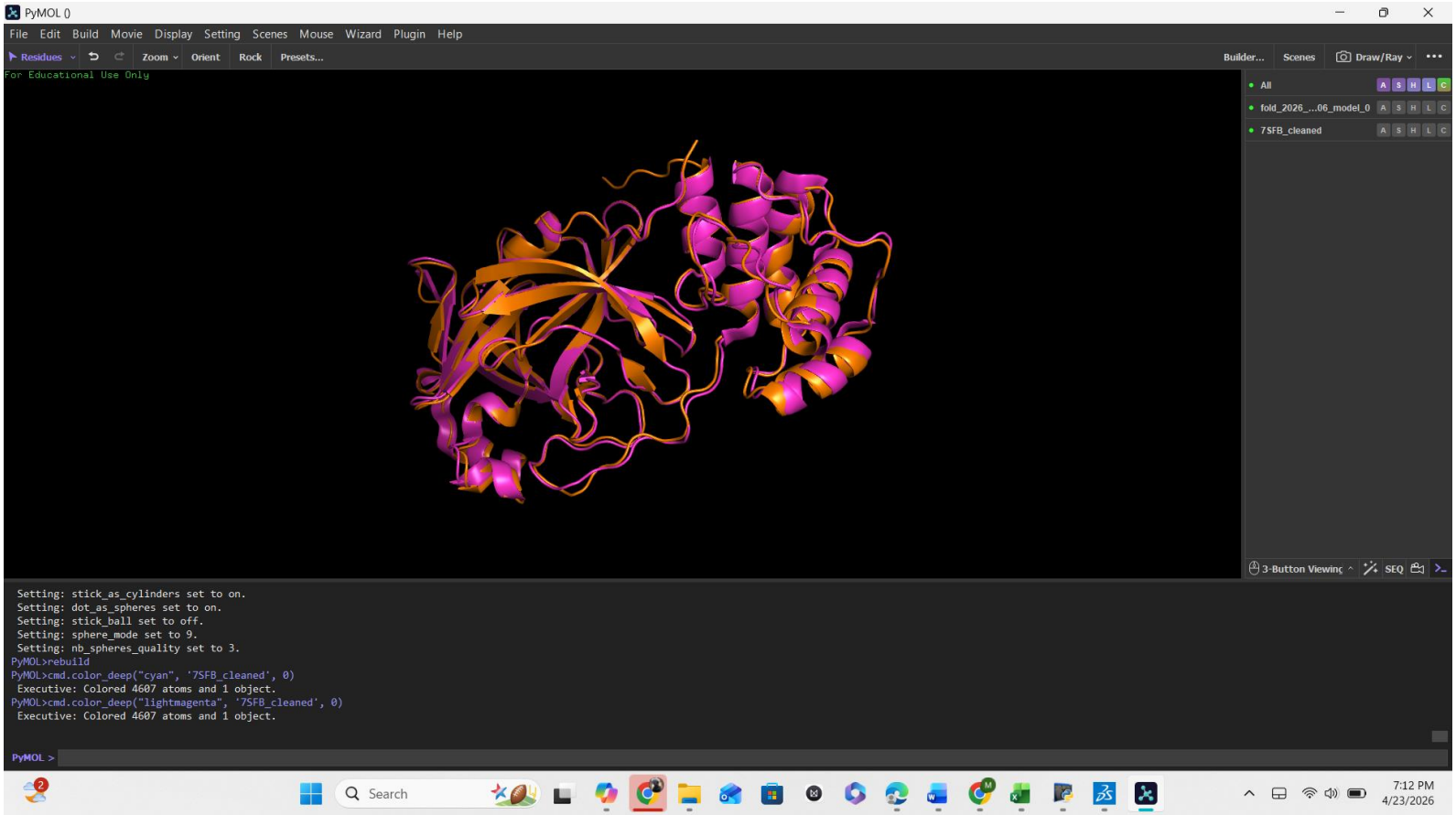


Figure 15 shows the alignment of the Alpha Fold 3 (AI) predicted protein structure overlayed on the real protein structure obtained in PDB and cleaned in chimera. RMSD value of 0.616 makes AI predicted structure high confidence and reliable

9. Results

9.1 Drug-Likeness Analysis

Drug-likeness evaluation revealed that nine out of ten compounds satisfied Lipinski's Rule of Five with no violations, indicating favorable pharmacokinetic properties. Lumefantrine exhibited two violations, suggesting potential limitations in oral bioavailability.

9.2 Molecular Docking Results

Docking analysis demonstrated that all compounds exhibited moderate binding affinity toward the target protein. Metformin showed the strongest interaction, followed by Diclofenac and Artemether.

Compound name	PubChem CID	Drug-likeliness (Lipinski rule of 5)	Binding Energy kcal/mol	Safety & ADMET	Balance Final Status
Aspirin	2244	Yes; 0 violation	-4.8	Low (High eye corrosion risk)	Not Approved
Ibuprofen	3672	Yes; 0 violation	-5.6	Moderate (High skin sensitization)	Approved
Acetaminophen	1983	Yes; 0 violation	-5.3	High (Very safe long-term use)	Approved
Naproxen	156391	Yes; 0 violation	-5.5	Moderate (Stable profile)	Approved
Diclofenac	3033	Yes; 0 violation	**-6.5**	**High (Most balanced profile)**	**Approved**
Caffeine	2519	Yes; 0 violation	-6	Moderate (Excellent BBB access)	Approved
Theophylline	2153	Yes; 0 violation	-4.7	Low (High drug interaction risk)	Not Approved

Metformin	4091	Yes; 0 violation	**-6.7**	**High (Strongest binding affinity)**	**Approved**
Artemether	68911	Yes; 0 violation	-6.3	Low (Dangerously high skin risk)	Not Approved
Lumefantrine	6437380	No; 2 violations	-6.1	Low (High skin risk/Poor absorption)	Not Approved

Discussion

The docking results demonstrated moderate binding affinity across all selected ligands, with binding energies ranging from −4.7 to −6.7 kcal/mol. Among the evaluated compounds, Metformin exhibited the strongest binding affinity, suggesting a favorable interaction within the active site of the protein. This enhanced binding may be attributed to its small molecular size and ability to form stable hydrogen bonds with key active site residues.

Diclofenac and Artemether also showed strong binding interactions, likely due to a combination of hydrophobic interactions and polar contacts that stabilize the ligand–protein complex. In contrast, compounds such as Aspirin and Theophylline exhibited weaker binding affinity, which may be associated with limited interaction capability and fewer functional groups contributing to binding.

The results further indicate that ligand binding is strongly influenced by structural properties such as polarity, functional groups, and molecular flexibility. Additionally, the consistency between docking results and drug-likeness evaluation highlights the importance of integrating multiple computational approaches.

Although Lumefantrine showed relatively strong binding affinity, its violation of Lipinski's Rule suggests potential limitations in pharmacokinetic performance. This emphasizes that strong binding alone is not sufficient for drug candidacy, and pharmacokinetic properties must also be considered.

The docking analysis was focused on the predicted active site region, which includes key residues responsible for ligand binding and enzymatic activity.

Conclusion:

This study demonstrates the effectiveness of integrating molecular docking, drug-likeness screening, and ADMET analysis in identifying potential drug candidates. Among all evaluated compounds, Metformin emerged as the most promising candidate due to its strong binding affinity and favorable pharmacokinetic profile. The findings emphasize that a multi-step computational approach provides more reliable insights compared to single-method analysis. These results provide a foundation for further experimental validation and future drug development studies.

The best ligand which balances three competing factors: Drug-likeliness (following Lipinski's Rules), ADMET Safety, and Binding Affinity (the strength of the drug-target interaction) is Metformin & Diclofenac with binding affinity of -6.7 to -6.5 respectively. It is the most balanced candidate among all in the data set which follows the Lipinski rule and has balanced ADME properties.